Yinghao He

Second-year PhD
in Computer Science
Purdue University

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Website
LinkedIn

Education

- **Purdue University** — *August 2025 - Present*
  Computer Science — West Lafayette
  – GPA 4.0/4

- **Tongji University** — *September 2021 - June 2025*
  Software Engineering — Shanghai China
  – GPA 4.88/5
  – Rank 4/228

Honors & Awards

- **National Scholarship** — *Year 2021-2022*
  The Highest Honor for Chinese Undergraduates (Top 1%) — The Ministry of Education
- **First-class Scholarship** — *Year 2022-2023*
  Scholarship for Distinguished Students(Top 3%) — Tongji University
- **Merit Student of Tongji University** — *Year 2021-2022*
  Outstanding Student Honor — Tongji University

Project Experience

- **Self-evolving Agentic Data Pipeline** — *May 2026 – Aug 2026*
  Analogy AI, Summer Research Intern — https://analogyai.ai/
  – **Challenge & Goal:** Collecting high-quality LLM training data at scale is costly, while general-purpose AI tools struggle with large, targeted requests. The project aimed to automate data sourcing, curation, and evaluation for domain-specific and frontier LLMs.
  – Developed sourcing and evaluation components of an agentic pipeline that converts natural-language data requests into curated datasets for LLM training.
  – Built agent-callable search tools across web and specialized data platforms, alongside an agentic crawler for permitted websites, to scale raw-data collection.
  – Designed an evaluation loop combining agent-based assessment with external validation to identify deviations from user requirements and guide iterative data curation.

Research Experience

- **Mitigating Sycophancy and Contextual Stubbornness in LLMs** — *May 2026 - Sept 2026*
  Analogyai & Purdue University
  – Developed CoPE-Bench to jointly evaluate sycophancy and contextual stubbornness across six conditions involving user pressure and contextual evidence.
  – Proposed SPAE, a training-free attention-steering method that uses model-identified token spans to suppress user pressure and amplify contextual evidence.
  – Evaluated SPAE across five backbones (4B–27B) against six baselines. On Qwen3.5-9B, reduced pressure following from 24.5% to 6.0% and improved contextual updating from 50.5% to 60.2%, with no accuracy loss on no-conflict controls.

- **Vector Database Research Project** — *Aug 2024 - Apr 2026*
  Purdue University
  – Designed and implemented a disk-based vector retrieval system using cluster-centroid indexing to support datasets exceeding memory capacity.
  – Developed a hybrid retrieval strategy combining compressed in-memory vector representations with multiple disk-reading methods.
  – Implemented estimated-distance filtering to exclude over 90% of candidate vectors before disk reads, reducing I/O while maintaining retrieval accuracy.

Publications

- **Suppressing Pressure, Amplifying Evidence: Self-Guided Attention Steering to Mitigate Sycophancy and Stubbornness** *Submitted to ICLR 2027*
  Yinghao He, Mengyu Xu, Haixiang Sun, Donghan Li, Yibo Wang, Lixu Wang, Kezhen Chen, Chi Li, Chunwei Liu, Bharat Bhargava, Chongyang Gao
- **DiskV: Disk-based Vector Search on Commodity Storage** *Submitted to SIGMOD 2027*
  Yunan Zhang, **Yinghao He**, Jianguo Wang
- **Reliable Replay through Spatial Coherence in Online Continual Learning** *Submitted to ICLR 2027*
  Haixiang Sun, Jiefu Zhang, **Yinghao He**, Yang Xu, Vaneet Aggarwal, Bharat Bhargava, Andrew Liu
- **FM-Bench: A Benchmark for Long-Horizon Management with Competing Agents** *Submitted to ICLR 2027*
  Tianyou Wang, Chongyang Gao, Kezhen Chen, Dong Chen, **Yinghao He**, Donghan Li, Wangcheng Xu, Yifan Xiao, Hongjiu Zhang, Chi Li

Technical Skills

- **Programming Languages**: C/C++, Python, Java, C#, Go, R